kader Academy

Machine Safety Specialist (CE)

Module 7: Start, stop and reset

TAB4: Risk reduction and standards
Tab4-7/9 20240816

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Tab4-7/9-Sh.2

Start, stop and reset

This module:

- EN-IEC 60204-1:2018 Electrical equipment of machinery
- EN-ISO 13850: 2016 Emergency stop devices

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Start

- The operator presses the start button to start the machine (an intentional operation with a control designed for this purpose)
  - By default, this also applies to restart or after setup. A machine should not start automatically when it is turned on.
- This must be done in a safe way. Is it manageable? (risk assessment, reset procedure, start warning, all security means 'OK', etc.)
- Automatic machines may start or restart themselves, but only if this does not cause dangerous situations (risk assessment).
- If there are several start buttons and operators may endanger each other, additional provisions must be in place to eliminate this risk.
- Tip: if available, check the C standard

Stop

- The operator presses the stop button to stop the machine(a control device, which must be present at which operating station, with which the machine can be stopped completely in a safe manner).
- The stop command takes precedence over the start command
- If the machine has come to a standstill, there are 2 options:
  - Normal shutdown: the power supply to the drives is interrupted
  - Operational shutdown: the shutdown state is monitored and maintained

EN IEC 60204-1:

Category 0: A non-controlled stop,
simple switching off the power supply (energy) to the drive.
(Safe Torque Off function of a frequency converter is also included in this)

Category 1: A controlled stop, drive provides fast (or appropriate) deceleration,
After that, the power supply (energy) is switched off.

Category 2: An operational stop, where the energy remains on the drive
when the machine is stationary.

Emergency stop is only allowed with a cat. 0 or cat. 1!

Tab4-7/9-Sh.5

Safety functions for frequency drives

- STO
  Safe Torque off, no more power to the engine, holding brake required with stored energy.
- SS1
  Safe Stop 1, drive brakes to a standstill, then STO, required at high mass inertia.
- SS2
  Safe Stop 2, drive brakes to a stop, then SOS. No brake required, no wear, machine can continue immediately after the stop.
- SOS
  Safe Operating Stop, motor is not switched off, is stationary but can still have torque, stays (regularly) in position, speed (0) is monitored. Reference between multiple coupled axes remains.
- SLS
  Safely Limited Speed, monitors exceeding one or more speed limits.

IEC 61800-5-2

Tab4-7/9-Sh.6

Safety functions for frequency drives

- If the speed is higher, STO or SS1 follows
- SSM
  Safe Speed monitor, output signal that is high when the speed is lower than the set limit.
- SDI
  Safe Direction, Monitors the direction of movement, activates a stop in the event of an error
  SLP
  Safely Limited Position, ensures that the position of an axle remains within specified limits, activates a stop in case of error.
- SBC
  Safe Brake Control, operates an external (holding) brake, always in combination with STO.
- SBT
  Safe Brake Test, Automatic test (control drive with dropped brake and monitor standstill) to test braking performance and wear.

IEC 61800-5-2

Emergency stop(function)

- Machinery must be equipped with emergency stop devices to prevent real or imminent dangerous situations.
- With the exception of portable machinery held by hand and machinery where the risk would not be reduced by an emergency stop device (risk assessment)
- Emergency stop devices are intended to support, not replace, other safety measures.
- The emergency stop function must be available and operational at all times, regardless of the operating mode.

The emergency stop function

- Must avert or reduce hazards to persons, damage to machinery or damage to workpieces;
- Triggered by a personal action (press button or pull emergency stop pull cord)
- Is dominant in all possible operating modes.
- Must not interfere with other safety functions

Emergency stop function

Design the emergency stop function in such a way that the operator does not have to consider the consequences of any emergency stop when deciding whether to use the emergency stop device.

Emergency stop function

Design the emergency stop function in such a way that, after operation of the emergency stop device, dangerous movements and processes are stopped in an appropriate manner, without the occurrence of hazards and without the intervention of any person, according to the risk assessment. Make sure that the stopping takes place in an appropriate manner.

To do this, apply the following principles:

- Choose an optimal deceleration time (braking time);
- Apply the correct stop category (category 0 or category 1, EN IEC 60204-1);
- If necessary, apply a predetermined shutdown sequence.

Emergency stop function

Select, assemble, connect and secure the components for the E-STOP function in such a way that they can perform their function under the expected environmental influences and operating conditions.

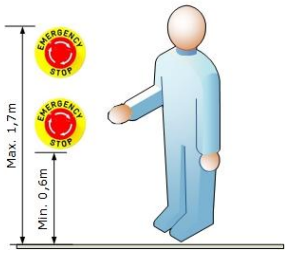
Keep in mind the:
- Frequency of operation and necessary periodic tests, e.g. in the event of irregular use;
- Influences due to e.g. shocks, temperature, vibrations, moisture, foreign bodies, liquids, corrosive materials.

Regularly check that the emergency stops are working properly and that the emergency stops have not been removed or taken out of operation.

Use SIL (EN IEC 62061) / PL (EN ISO 13849-1)
In accordance with EN ISO 13850, the E-STOP function must be implemented at least PLc or SIL1.

Emergency stop device lopcations

- At any operating position, except if the risk assessment indicates that this is not necessary;
- In other locations, if the risk assessment shows that this makes sense as:
  - Inputs and outputs
  - Positions where intervention takes place (with hold to run control)
  - Positions where human-machine interaction is to be expected such as loading/unloading
- Emergency stop device must be easy and safe to operate;
- Hand-operated devices at heights between 0.6 and 1.7 meters.



Emergency stop devices

- Red with a yellow background.
- Locks itself, unlock on the spot.
- Equipped with forced switching NC contacts.
- Easy to reach / operate.

Emergency stop devices

- Do not use text or characters on an emergency stop button (should).
- If a symbol is needed:
- If it is necessary to indicate the release direction, this must be in (approximately) the same color as the actuator to avoid confusion with the operating direction.

Unlock

- Persoonlijke actie;
- Vanaf de plaats waar de noodstop bediend is;
- Machine start niet op door ontgrendelen;
- Na ontgrendelen is herstarten van de machine (aparte actie) toegestaan.

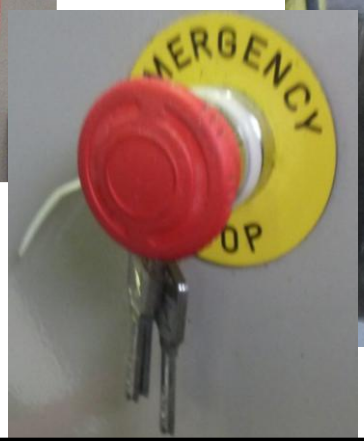
Voor ontgrendelen van de noodstop moet de machine worden geïnspecteerd: 'waarom is de noodstop gedrukt?'
Neem hiervoor instructies op in de gebruiksaanwijzing.

Gebruik geen noodstoptoestel dat met een sleutel moet worden ontgrendeld (should).
Verwijder in ieder geval de sleutel na ontgrendelen (gebruiksaanwijzing)

Unintentional operation

- Prevent accidental operation by design.
- Choose a different location.
- Select another type, size, shape, etc.
- Preferably no collar, unless other measures are not practical
- A 'collar' must not lead to injuries
- If an emergency push button is intended to be operated with the palm of the hand, this should still be possible after a measure against unintentional operation has been placed. From all foreseeable positions of the operator.

Unintentional operation



Emergency stop devices

If emergency stop buttons are fitted to detachable or wireless control units, at least one emergency stop button must be permanently mounted on the machine.

In addition, take measures to avoid confusion with inactive control units, such as:

- Colour changing;
- Shielding inactive ES;
- Storage of unused control units.

Explain the measures in the instructions for use.

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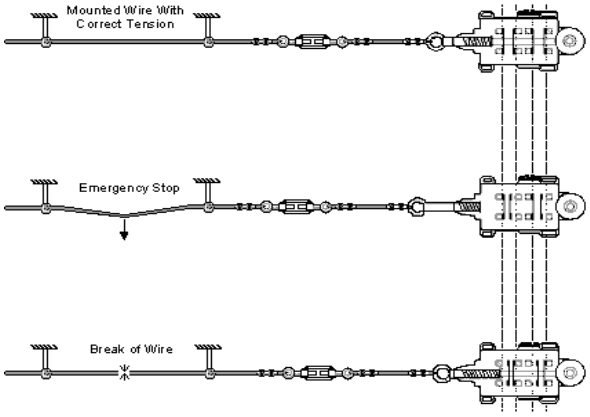
Emergency stop devices

Tab4-7/9-Sh.20

Emergency stop cord

Measures to prevent hazards due to rupture or disconnection of the emergency stop rope.



Emergency stop cord

Please note:
- Maximum elongation and force required to generate the emergency stop;
- Minimum distance between the cord/cable and the nearest object;
- Visibility of the cable;
- If possible, the reset should be placed in such a way that the entire length of the cord/cable is visible.

Include in the owner's manual that after operation and before resetting, the machine must be checked along the entire length of the emergency stop cord to determine the cause of the activation.

Main switch as an emergency device

nooduitschakeltoestel
handmatig bediend toestel dat wordt gebruikt om bij risico van elektrische schok of een ander risico van elektrische oorsprong, de elektrische voeding naar de installatie of een deel daarvan te onderbreken

- In some cases, a main switch (switch-disconnector) can be used as an 'emergency switch'.
- Easily accessible for the operating staff.
- Only for stop category 0.
- Control panel red, background yellow.
- Not in combination with other E-STOP device (confusion).
  EN IEC 60204-1

Tab4-7/9-Sh.23

Main switch as an emergency device

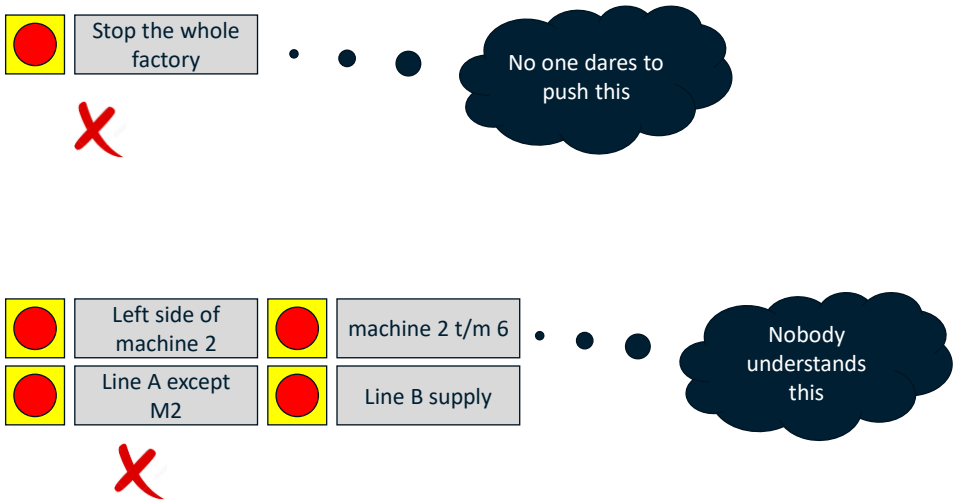
- Name a machine / situation / calamity in which it is not desirable that the main power supply is accidentally switched off instead of the emergency stop function is activated?

Tab4-7/9-Sh.24

Main switch as an emergency device

- Name a machine / situation / calamity where the use of an emergency shutdown switch could be an acceptable solution?

Span of control, too easy and too difficult



Span of control

- Choose 'single span of control' so that each emergency stop device stops the entire machine. (shall)
- In exceptional cases, multiple spans of control are used, for example in the event of unnecessary disruption to production or additional hazards.
- Span of control can therefore be
  - Section
  - Machine
  - Group of machines (line)

Span of control

Arguments for granting a span of control:

- Layout and visibility;
- Ability to recognize hazards (sight, sound, smell);
- The production process;
- Foreseeable exposure to hazards;
- Adjacent Hazards.

Span of control

In multiple spans of control:

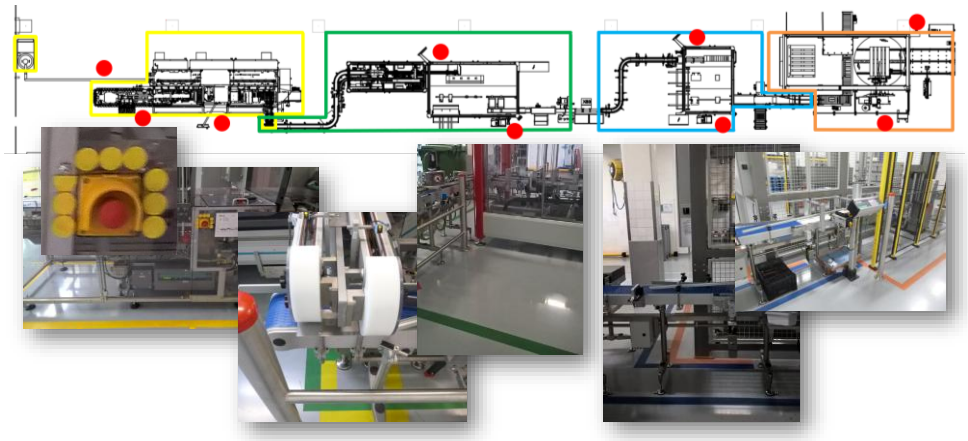
- Clearly defined and recognizable;
- Clear link between a hazard and the emergency stop pusher;
- The span of control is clearly recognizable in the emergency braking device;
- Pressing an emergency stop does not create additional hazards in a different span of control;
- Pressing an emergency stop does not prevent the operation of an emergency stop in another span of control;
- Include instructions in the instructions for use.

Do not install E-STOP pushers with different spans of control next to each other (if practically possible).

Tab4-7/9-Sh.29

Span of control

An example of making emergency stop zoning visible (does not work in a sand-lime brick factory….)



Tab4-7/9-Sh.30

Maintenance

The adjustment and maintenance points must be located outside the danger zones. It must be possible to adjust, maintain, repair and clean when the machine has come to a standstill.

If one or more of the above conditions cannot be met for technical reasons, measures must be taken to ensure that these operations can be carried out safely (see section 1.2.5 of the Machinery Directive).

Machinery Directive Annex1 § 1.2.5

If the machine is to be able to operate for certain operations with a relocated or removed guard and/or a protective device switched off, the function selector switch for the operating or operating mode must be activated at the same time:

1. Disable all other operating or operating modes;
2. Enable the operation of hazardous functions only by means of controls which must be operated continuously;
3. Enable the operation of hazardous functions only in conditions of reduced risk, thereby preventing any hazard arising from interconnected controls;
4. To prevent the operation of hazardous functions due to intentional or unintended influence on the machine's sensors.

If these four conditions cannot be met **simultaneously**, the function selector must activate other protective devices designed and constructed to ensure a safe workspace.

In addition, the operator must be able to control the operation of the components **under his influence from the control station**.

Working on switched off Machines

In order to work on a machine, it must be switched off.

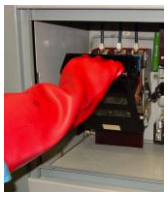
EN IEC 60204-1 §5.4
A device must be fitted to prevent unexpected start-up when switching on the machine or part thereof may cause a hazard (e.g. during maintenance).

These devices must be suitable for their intended use, must be clearly and accessible and clearly identifiable in view of their function and purpose.

Where their function and purpose are not self-explanatory (e.g. because of their location), these devices should be marked to indicate which part of the machine is affected.

Device to prevent unexpected switch-on (work switch)

Examples:
- Lockable mains separator,
- Remove fuses,
- Motor protection switch with interlocking,
- Removable connections.



Demands:
- Convenient and practical for intended use,
- Applied in a suitable place,
- Clearly recognizable which part they disable,
- Equipped with suitable tool that prevents accidental closing.

Device to prevent unexpected switch-on (work switch)

(EN-IEC 60204-1 § 5.4)
Devices that do not perform an isolating function (e.g. a contactor that is switched off by means of a control circuit, or an electronic drive system with the function of safely switched off torque (STO) according to IEC 61800-5-2) may only be used to prevent unexpected switching on during activities such as:

- Inspecties;
- Instellingen;
- Work on the electrical equipment where:
  - there is no risk of electric shock (see chapter 6) and risk of burns;
  - The shutdown remains in effect during the entire work;
  - It concerns minor work (e.g. the replacement of plug-in units where existing wiring remains unchanged).

We call this an 'operating switch in the control current'.

Reset

A reset is the manual function that resets all safety functions (e.g. emergency stop button, function lock on entrance doors, light curtains, etc.) before a restart is possible.

Examples where a reset feature is NOT a safety feature:

- After closing a function-blocking shield that consists of a hood or small door that no one fits behind. It is inconceivable that anyone is still in the machine when the hood is closed.
- After resetting an emergency stop button, because resetting the emergency stop function is already done by unlocking the button.
- After walking away from a scanner area that has been chosen in such a way that it is impossible for someone to be in the range without being detected.

EN ISO 13850

When the emergency stop function is activated:

- It shall be maintained until it is manually reset;
- It shall not be possible for any start command to be effective on those operations stopped by the initiation of the emergency stop function.

The emergency stop function shall be reset by intentional human action.

Resetting of the emergency stop function shall be operated by disengagement of an emergency stop device

The reset shall not initiate machine start up.

- NOTE The emergency stop function cannot be considered as measure of prevention of unexpected start up as described in ISO 12100.

EN ISO 13855

Is a reset function needed?

- Check whether the determined position allows persons to remain between the safeguard and the hazard zone without being detected. In this case, supplementary measures will be required depending on an additional risk assessment.
- NOTE 3 An example of a supplementary measure is a manual reset switch positioned outside the hazard zone and the space between the safeguard and the hazard zone. Its position is selected to allow someone operating it to readily check that no one is within the hazard zone or in the space between the safeguard and the hazard zone. For the requirements of a manual reset function, see ISO 13849-1:2006, 5.2.2.

The reset button

- Is placed on the safe side of a light curtain or shield;
- provides a good view of the danger zone. If this is not possible, a special reset procedure is required (such as reset in the danger zone with button 1 and then reset outside the zone with button 2 within a certain time);

Tab4-7/9-Sh.39

The reset function (as a safety function)

- is set in motion by (deliberately) pressing a separate manual button;
- is only implemented when all protections are operational;
- does not cause dangerous situations (nothing will move);
- prepares the control for a start signal;
- is only carried out by 'monitored signal change'
- has a SIL or PL level that may be different from the safety functions of the light curtain or the function-blocking shield..

Tab4-7/9-Sh.40

EN ISO 13849-1:2023

5.2.2.3 Manual reset function

The re-establishment of the safety function by resetting of the safeguard cancels the stop command. If indicated by the risk assessment, this cancellation of the stop command shall be confirmed by a manual, separate and intended action (manual reset).

The manual reset function shall:

— be provided through a separate and manually operated device that is separate from start command,

— only be achieved if all affected safety functions and safeguards are operational,

— not initiate a hazardous situation by itself,

— be initiated by intended action,

— enable the control system to accept a separate start command, and

— only be accepted by monitored signal change in order to avoid foreseeable misuse.

Operator lockout

As an alternative to a reset procedure

- The operator opens a function-blocking access door, the machine stops.
- Before the operator walks in, he/she hangs his/her own padlock on the door.
- The door can then no longer be closed (completely), restart is impossible due to the function block.
- If there are x people inside, there are also x locks on the door (use hasps if necessary).
- When all padlocks are gone, the door can be closed again and there is no one left in the danger zone. The machine may start again, a reset function is not necessary.

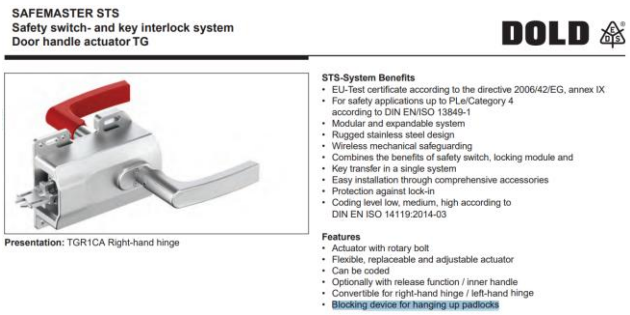
Operator lockout

Advantage:

- Placing a padlock is for your own safety (as opposed to reset function)

Disadvantages:

- Confusion with LOTOTO (no separators)
- Confusion in simultaneous reset and operator lockout procedures



SAFEMASTER STS
Safety switch- and key interlock system
Door handle actuator TG

DOLD

Presentation: TGR1CA Right-hand hinge

STS-System Benefits
- EU-Test certificate according to the directive 2006/42/EG, annex IX
- For safety applications up to PLe/Category 4 according to DIN EN/ISO 13849-1
- Modular and expandable system
- Rugged stainless steel design
- Wireless mechanical safeguarding
- Combines the benefits of safety switch, locking module and
- Key transfer in a single system
- Easy installation through comprehensive accessories
- Protection against lock-in
- Coding level low, medium, high according to DIN EN ISO 14119:2014-03

Features
- Actuator with rotary bolt
- Flexible, replaceable and adjustable actuator
- Can be coded
- Optionally with release function / inner handle
- Convertible for right-hand hinge / left-hand hinge
- Blocking device for hanging up padlocks

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